

Project Overview

[PolyU's Education 4.0 initiative](#) is a university-wide strategy to transform teaching and learning into a student-centred, AI-empowered, and personalised experience. It combines AI and digital transformation to move beyond one-size-fits-all lectures that help students take ownership of their learning and develop future-ready skills.

AI has turned coding from a specialist skill into a commodity. A growing majority of developers now rely on AI coding agents in their regular work, and a substantial share of people building real tools with AI have no formal computer science training. For people without formal computer science and programming training, AI coding tools lower the barrier to turning ideas into working software and iterate quickly without first mastering syntax, tooling, and frameworks. For computer science graduates, the rise of AI shifts the skill set that matters: value moves from code writing to framing problems, designing systems, specifying requirements, evaluating and refining AI output, and orchestrating agentic workflows.

The objectives of this project are to develop a GenAI-powered learning platform with learning contents and activities.

- Develop mini-courses with interactive learning contents and activities to makes the learning of software development and engineering more interactive, engaging and fun.
 - The following topics should be covered: Fundamentals of software engineering and development, effective use of AI coding agents in software development, soft and product skills in software development, etc.
 - Formats of learning content and activities: text and images, interactive simulations, interactive role play or mock interviews, games/mini-games, AI-narrated animations and slides, online coding, etc.
 - Use AI to teach the learners and provide a personalized learning and tailored learning content to engage students from different backgrounds and learning goals. For instance, the content should be accessible to non-Computer Science majors who want to understand software development and software engineering, as well as to Computer Science graduates preparing for careers in related fields in the age of AI.
- Instructors should be able to customize (manually or using AI) the learning contents and activities of the existing mini-courses. They can also use AI to develop new mini-courses by inputting their own topics, requirements, and teaching content, and refining learning content and activities. They can import their students into their courses and track students' participation and engagement in learning activities.
- Allow the user to generate mini-courses for other subjects/domains and let learners generate their own topics and path for self-learning.

Each group is expected to create multiple prototypes for brainstorming and validation, but should work collaboratively on one final app with integrated features.

- Each team member should demonstrate sufficient contribution to the project (for example, by developing different features and integrating them into the final app).
- The app should include a good variety of features with sufficient complexity so that different team members can collaborate meaningfully on its development.

In the project, you may also use GenAI for various tasks, such as brainstorming ideas, design, evaluating alternatives, development and coding, and testing. You are welcome to use any programming language, framework, or library. In the age of AI, knowing what to build and why is more important than ever: AI can rapidly generate code and features, but it cannot replace human judgment about user needs and problem framing.

- Try to find more inspiration from social media (e.g., Reddit, x, YouTube) on what people are developing, the emerging trends, and the latest developments in the field.

- You should meet with stakeholders and users to collect requirements and feedback on your prototype, and explain how the prototypes can be used for iterative improvement.
- Try to understand how AI implements the solution, system architecture, and learn new skills throughout the project.

Anthropic's defines "AI Fluency" as the ability to collaborate with AI systems effectively, efficiently, and responsibly. Its 4D framework breaks this into four competencies: Delegation (deciding what to hand to AI), Description (clearly specifying goals, constraints, and process), Discernment (critically evaluating AI outputs and reasoning), and Diligence (verifying, documenting, and taking accountability for results). In this project, each team should study the AI Fluency framework, explain how it can be applied to your project and share these reflections in your final presentation.

Group Formation

Students should form a group of 4 members by completing the MS form by the end of week 3 (i.e 31/1/2024 (SAT)). Students without a group will be randomly assigned a group after the deadline.

Each group should create a public GitHub Repository to store your code, documents and deliverables.

- Each team member should use the GitHub repository (e.g., issues, project board, etc) to manage and collaborate on your project.
- Your project source code and other related files (e.g., documentation) should be uploaded to your project repository in the course project's GitHub organization.
- All students in the group should actively participate during the software development (E.g., coding, testing, writing documentation), commit actively to the repository and adopt good software engineering practices.

The project meeting will be held during the lectures in week 8 or 9. Each group will have around 5-10 minutes to discuss its project progress. You should prepare a 1–2-page summary of your progress, division of work, and your system design/prototype for discussion. We may also clarify the requirements in the meeting.

Week 11: Final Submission

Deliverables

- The prototype should be deployed to some publicly accessible platform (e.g. Vercel)
- A 5–8-minute overview video of your system and how to use the platform
- Your presentation slides and a final report
 - Overview of your project, system design and features
 - Discuss your software development process (requirement, design, development, testing, deployment, etc) and how your team collaborates in your project
 - User manual for admin, instructor and students and links to demo videos of the features

Week 12-13: Presentation

The presentation will be held in week 12 and 13. Each group will have max 8 minutes (to be confirmed) to present their project and around 5 minutes for QA.

The presentation slides should cover the following aspects:

- An overview of your project, features, and system architecture.
- The demo and features of your prototype/system (You may show part of your demo video)
- Sharing findings and reflection on the project. Examples:
 - o Development Process
 - Briefly introduce your team's technical/programming background and discuss your development process and examples of how your group collaborates. Discuss how each team member is contributing to the project and the challenges encountered (if any)
 - How did you improve the features, UI and UX of the platform? Which feature in the project was the most challenging to implement, and how did you solve the problem?
 - o Use of AI in software development
 - Which parts of the code are generated by AI and which parts are manually coded? How is your team collaborating while using GenAI to complete the software development tasks? What works well/not so well?
 - How are you using Coding agents and AI to support the software testing and perform other tasks in software development? Showcase your examples, practices and tips when using coding agents for software development.
 - Share how your team applied Anthropic's AI Fluency framework in your workflow. Is it useful to you? Provide concrete examples of how it shaped your project implementation.
 - o Overall reflection: Reflect on what you have learnt/the findings from the project, what you have done well/not done so well, what the challenges and solution, etc.
- Appendix: You may include other information/screenshots that are relevant to your project.

Requirements:

- AI-generated slides are not preferred. Do not directly copy and paste AI output to the report or slides. Do not include content that you have not covered or concepts that you cannot explain. Try to incorporate your own style, tone, and human touch.
- Try to remove the "AI smells" and obvious signs of unpolished/unedited AI-generated contents from your submitted deliverables, such as the app itself, the report, and presentation slides
- You should not read the scripts from mobile phone during the presentation. Try to put more effort on delivering a good and engaging presentation (use AI to help you improve your presentation contents and delivery)
- Advances in AI have made it easier to generate prototypes quickly, and it is becoming more difficult to assess students' performance by watching their demos or presentations alone. In your submission and presentation, you should show in more details your project process and supporting evidence of your effort and contribution (e.g. photos or videos from user interviews, how you test your software, how you develop the deliverables with AI and what you contributes, etc).

Evaluation

- Project discussion meeting (5%)
- Final presentation and deliverables (85%)
 - System features and prototype (40%):
 - Presentation, Sharing, findings, and reflection (45%)
- Peer evaluation (10%)
 - Each group should
 - Try the platforms of two other groups assigned to you
 - Attend the presentation of your two assigned teams and ask at least one question
 - Evaluate your peers' platform/presentation and provide feedback.

Rubrics (Tentative)

Criteria	Good (3.5-4)	Satisfactory (2.5-3)	Needs Improvement (1-2)
Project Discussion Meeting (5%)	Clear progress summary with a well-defined prototype and division of work. Team shows active participation and a strong grasp of requirements.	Progress shared but limited details on the prototype or task division. Participation is evident but uneven among members.	Lacks clarity on progress or work division. Prototype/design unclear. Some members appear disengaged.
System Features & Prototype (35%)	Features are innovative/tailored to the needs of the stakeholders. Clear articulation of motivations, differentiation from existing tools. GenAI features are well integrated. The prototype is functional, responsive, and deployed with a good UI/UX. Implemented unique/unexpected features which stand out from the other teams	Features are implemented but reflect more basic or imitative functions . Motivation and differentiation are partly explained. Prototype deployed but with limited interactivity or UI/UX polish.	Features are very basic or similar to existing tools. Lack of motivation or justification. Prototype incomplete, not responsive, or not deployed.
Presentation, Sharing on project process, and reflection (45%)	Clear, engaging, professional delivery with confident pronunciation, varied tone, and smooth demo. The slides are well-structured, with clear visuals that enhance understanding. Provides in-depth reflections and discussion of your project progress, including the development process, team collaboration, and what you/AI contributed individually or collaboratively (successes & challenges). Insights are thoughtful and specific. Clear takeaways about the learning process.	Presentation is understandable but less polished. Delivery may include minor clarity/pronunciation issues; slides/content may be somewhat repetitive or cluttered. Audience engagement is limited. Project process and reflections are present but general or brief . The development process and GenAI usage are described superficially. Limited insights about design choices, challenges, or lessons learned.	Presentation unclear, monotonous, or poorly structured . Slides confusing with missing sections. Demo ineffective or absent. Audience disengaged. Reflection minimal or missing . No real insights beyond surface-level description. Lacks discussion of GenAI usage, challenges, or lessons learned.

Links

- [AI 時代軟體必修課](#)
- GPTutor
 - o <https://gptutor.comp.polyu.edu.hk/user-guide>
- EvoClass
 - o <https://www.evoclass.ai/>
- Teach me now
 - o <https://teachmenow.ai/>
- Start.dev
 - o <https://start.dev/>
- AI Fluency
 - o <https://academy.claude.com/courses/ai-fluency-for-builders>
 - o <https://academy.claude.com/collections/ai-fluency>
- [I Can Spot AI Writing in Seconds \(4 Signs That Make It Obvious\)](#)
- [AI簡報術最終回 這不是AI簡報教學 而是不可被取代的簡報思維](#)